Adaptive Gait Generation and Postural Stabilization Framework for Quadruped Robots via IMU Feedback Integration

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***Abstract*—This paper presents a lightweight, computationally efficient motion control framework for articulated quadruped robots that achieves stable blind locomotion using exclusively proprioceptive inertial feedback. A strictly decoupled dual pipeline control architecture isolates rhythmic gait generation from reactive postural stabilization, enabling independent tuning and fault tolerant operation. Foot trajectories are shaped using quintic polynomial interpolation functions $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ that enforce zero velocity and zero acceleration at boundary conditions, eliminating mechanical impact transients. A multistage signal conditioning pipeline incorporating exponential moving average (EMA) prefiltering, configurable deadband suppression, gain scheduled PID with back calculation antiwindup, and cascaded lowpass filtering produces smooth corrective outputs free of high frequency jitter. During dynamic locomotion, phase aware rotational compensation matrices $R_y(\theta)R_x(\phi)$ are selectively applied to stance phase limbs via real time foot altitude thresholding, while swing phase limbs maintain unmodified trajectories to preserve stride clearance. Simulation based validation confirms the system's capacity to reject aggressive lateral disturbances while maintaining the Zero Moment Point (ZMP) within the support polygon.**

I. Introduction

Legged robots possess the innate structural capacity to navigate discontinuous terrain and extreme topographies that inherently challenge conventional wheeled platforms. However, developing resilient locomotion policies capable of stabilizing a heavily articulated system with multiple degrees of freedom against unmodeled disturbances remains a rigorous control challenge.

Contemporary solutions rely heavily on computationally intensive algorithms, such as Model Predictive Control (MPC) and Reinforcement Learning (RL), to estimate optimal contact dynamics [1], [2]. While highly performant, these stochastic solvers predominantly require specialized hardware accelerators or processors operating at high frequencies, severely precluding deployment upon the embedded edge devices ubiquitous in modern mechatronics.

Conversely, Central Pattern Generators (CPGs) driven by heuristic rules and paired with reactive reflex feedback loops can approximate biological stability at a fraction of the computational cost. When inertial telemetry of high fidelity, specifically instantaneous roll and pitch orientation [3], is directly integrated into kinematic calculations operating in real time, robust blind ambulation becomes tractable [4].

This work proposes four key contributions that distinguish it from conventional CPG based quadruped controllers:

1) A **quintic polynomial trajectory planner** utilizing the normalized interpolation function $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ to enforce zero velocity and zero acceleration boundary conditions at both liftoff and touchdown, eliminating mechanical impact transients that corrupt inertial sensor baselines.
2) A **multistage signal conditioning pipeline** comprising: (a) exponential moving average (EMA) prefiltering, (b) configurable deadband suppression, (c) gain scheduled PID with integral leakage and back calculation antiwindup [5], and (d) cascaded lowpass filtering, producing smooth corrective outputs free of servo jitter.
3) A **phase aware stance correction** mechanism that applies rotational compensation matrices $R_y(\theta)R_x(\phi)$ exclusively to limbs classified as load bearing via foot altitude thresholding in real time ($|Z - Z_{stance}| \leq Z_{thresh}$), while swing phase limbs maintain unmodified trajectories.
4) A **strictly decoupled dual node architecture** that isolates rhythmic gait generation from reactive postural stabilization, enabling independent tuning, independent failure modes, and asynchronous execution at differing control frequencies.

II. System Architecture

A. Node Topology

The framework operates as a distributed processing architecture partitioned across two functionally isolated computational nodes. The primary *Gait Generator* node manages all rhythmic trajectory computation, inverse kinematics resolution, and actuator command dispatch. An asynchronous *Balance Controller* node exclusively handles inertial measurement acquisition, signal conditioning, and PID correction computation. Communication between the two nodes is strictly limited to lightweight corrective vector messages. This architectural segregation guarantees that trajectory generation anomalies cannot preempt or block critical sensory compensations, and vice versa.

B. State Representation

Global governance overlying the control architecture is dictated by boolean gating, handling two discrete operating states:

- **HOME:** A static configuration lacking cyclic translation vectors. The controller computes uniform adjustments in joint space rigidly mapped to a resting stance. Base angular deviations translate proportionally into antagonistic torque compensations aimed at continuous postural rectification. A configurable startup delay of T_{delay} seconds prevents premature correction during system initialization transients.

- **GAIT:** The dynamic state active during commanded planar translations. The platform executes cyclic polynomial trajectories using configurable phase shift arrays to generate trot ($\phi = [0, 0, 0.5, 0.5]$), walk ($\phi = [0, 0.25, 0.5, 0.75]$), and other gaits involving multiple limbs. Spatial corrections derived from inertial feedback are applied exclusively to limbs in stance phase, while swing legs maintain zero compensation to ensure uninhibited stride clearance.

III. Kinematics and Gait Planning

A. Inverse Kinematics

To mandate the endpoint position (x, y, z) of each limb relative to the geometric origin of the base frame, a nonlinear inverse kinematics (IK) model [6] derives the required articular angles $(\theta_1, \theta_2, \theta_3)$ for the hip, thigh, and calf joints. The physical topology structuring the Cartesian workspace includes a platform length of $L = 250$ mm, a platform width of $W = 193$ mm, and individual linkage spans $l_1 = 45$ mm, $l_2 = 107$ mm, and $l_3 = 116$ mm.

The mathematical solution avoids geometric singularities by enforcing strictly restricted bounding limits calculated dynamically during system initialization. Fig. 1 illustrates the reference frame assignments at each joint origin, with RGB axes denoting the local coordinate systems used for forward and inverse kinematic resolution.

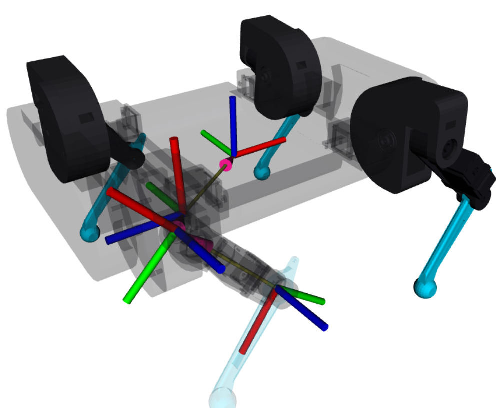


Fig. 1: *Simulation model with reference frames visualized at each joint. Each joint origin displays RGB axis indicators (red=x, green=y, blue=z) used for geometric IK resolution across the three DOF limb topology.*

B. Quintic Trajectory Formulation

Minimizing ground impact velocity is mathematically imperative; violent mechanical shocks corrupt inertial sensor baselines and induce structural fatigue [7]. Therefore, foot trajectories are shaped using the normalized quintic interpolation function:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5 \tag{1}$$

where $\tau = t/T \in [0, 1]$ and $s \in [0, 1]$. This function enforces the critical boundary conditions:

$$\begin{aligned} s(0) &= 0, \quad s(1) = 1 \\ \dot{s}(0) &= \dot{s}(1) = 0 \\ \ddot{s}(0) &= \ddot{s}(1) = 0 \end{aligned} \tag{2}$$

The resulting velocity and acceleration profiles are obtained by differentiation:

$$\dot{s}(\tau) = \frac{1}{T}(30\tau^2 - 60\tau^3 + 30\tau^4) \tag{3}$$

$$\ddot{s}(\tau) = \frac{1}{T^2}(60\tau - 180\tau^2 + 120\tau^3) \tag{4}$$

Fig. 3a visualizes the position, velocity, and acceleration profiles, confirming the enforced zero derivative boundary conditions.

For the swing phase of duration $T_{swing} = 70$ frames, the foot transitions from liftoff coordinate **A** to touchdown coordinate **D** along a D shaped parabolic arc with configurable lift height $h_{lift} = 40$ mm:

$$\mathbf{p}_{swing}(\tau) = \mathbf{A} + s(\tau)(\mathbf{D} - \mathbf{A}) \tag{5}$$

$$z(\tau) = z_{ground} + 4h_{lift} \cdot s(\tau)(1 - s(\tau)) \tag{6}$$

The resulting D shaped foot path is illustrated in Fig. 2, showing the swing arc and ground retrace phases with annotated liftoff and touchdown boundaries.

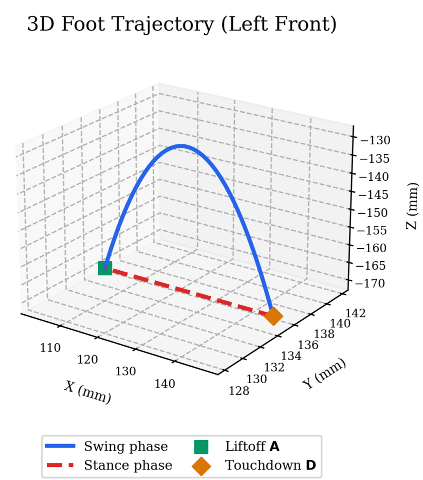
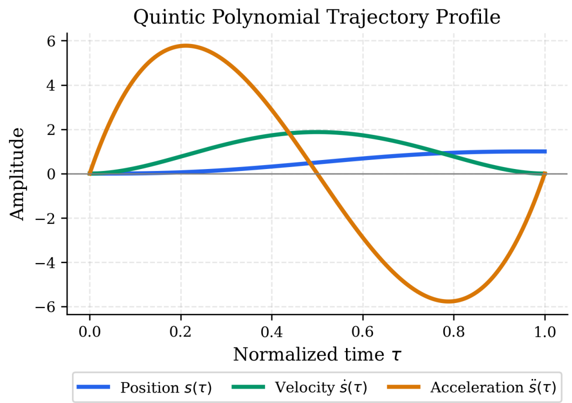


Fig. 2: *3D foot trajectory for the left front limb. The swing phase (blue) traces a quintic parabolic arc from liftoff* **A** *to touchdown* **D***, while the stance phase (red dashed) retraces along the ground plane at $z = -170$ mm. Stride length $L_{stride} = 45$ mm, lift height $h_{lift} = 40$ mm.*

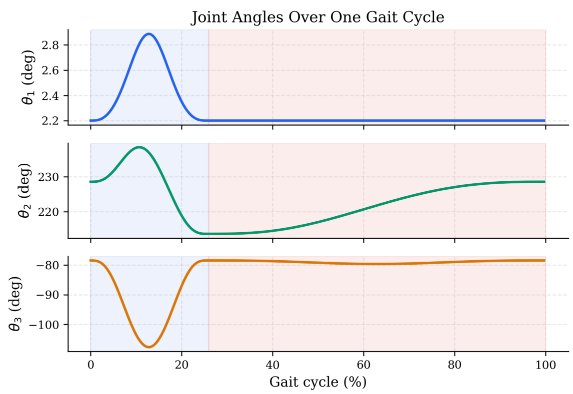
During the stance phase of duration $T_{stance} = 200$ frames, the foot retraces along the ground plane using the same quintic interpolation to ensure smooth velocity profiles in the reverse direction. The stride length is set at $L_{stride} = 45$ mm, centered about each leg's neutral x_{center} position.

Overlapping gaits involving multiple limbs are generated by applying phase shift offsets ϕ_i to the trajectory index for each limb. For a standard trot, diagonal pairs share identical phase offsets $\phi = [0.0, 0.0, 0.5, 0.5]$, producing the characteristic diagonal synchronization with two beats.

Fig. 3b presents the resolved joint angles θ_1, θ_2, θ_3 for the left front limb over one complete gait cycle, with swing and stance phases explicitly shaded.



(a) *Quintic polynomial PVA profiles.*



(b) *Resolved joint angles for the left front limb.*

Fig. 3: *Trajectory generation and inverse kinematic mapping. (a) The normalized quintic position, velocity, and acceleration functions dictating foot motion. (b) The resulting joint angle solution θ_i over one full gait cycle, with swing (blue, 0–26%) and stance (red, 26–100%) phases highlighted.*

IV. IMU Based Postural Stabilization

Inertial corrections are computed asynchronously within the strictly isolated Balance Controller. Raw quaternion orientation data is converted to Euler roll (ϕ) and pitch (θ) angles [3] and processed through a multistage transformation pipeline.

A. Static Home Posture Correction

During the static home phase, all four limbs maintain sustained ground contact. The correction pipeline proceeds in five sequential stages:

Stage 1: EMA Prefiltering. Raw inertial measurements are smoothed via an exponential moving average of the first order to suppress sensor noise at high frequencies:

$$\begin{aligned} \hat{m}_k &= \alpha_{raw}\hat{m}_{k-1} + (1-\alpha_{raw})m_k \\ \alpha_{raw} &= 0.4 \end{aligned} \tag{7}$$

Stage 2: Deadband Suppression. A configurable deadband of ±0.02 rad eliminates trivial oscillations that would otherwise generate unnecessary servo activity:

$$e_{dz} = \begin{cases} e-d & e>d \\ 0 & |e|\le d \\ e+d & e<-d \end{cases}, \quad d = 0.02 \tag{8}$$

Stage 3: Gain Scheduled PID with Antiwindup. A full PID controller [5] with three novel enhancements processes the deadband filtered error:

$$\begin{aligned} u_{PID} &= K_P \cdot g_s \cdot e \\ &\quad + K_I \cdot g_s \cdot \mathcal{I} + K_D \cdot \dot{e}_{lpf} \end{aligned} \tag{9}$$

where the gain schedule factor $g_s \in [0.5, 1.0]$ scales proportionally to the error magnitude, preventing corrections that are overly aggressive near zero:

$$g_s = \text{clamp}\left(\frac{|e|}{e_{thresh}},\ 0.5,\ 1.0\right) \tag{10}$$

where $e_{thresh} = 0.08$ rad.

The integral accumulator features exponential leakage decay ($\lambda = 0.9995$) to prevent windup at steady state, bounded by symmetric saturation limits ($|\mathcal{I}| \le 1.5$). Additionally, back calculation antiwindup reduces the accumulator when the PID output exceeds saturation bounds:

$$\mathcal{I}_k = \mathcal{I}_k - \frac{u_{raw} - u_{sat}}{2K_I} \quad \text{if} \quad |u_{raw}| > u_{max} \tag{11}$$

The derivative term is presmoothed with a dedicated LPF ($\alpha_d = 0.7$) and includes a derivative ramp threshold ($e_{ramp} = 0.005$) that zeroes the derivative when the error is negligibly small, preventing derivative kick.

Stage 4: Output Saturation. The composite PID output is clamped to $|u| \le 1.5$ rad to protect actuator limits.

Stage 5: Output Lowpass Filtering. A final lowpass filter of the first order smooths the corrective output to eliminate residual step discontinuities:

$$\begin{aligned} H_k &= \alpha_c H_{k-1} + (1-\alpha_c)\, u_{sat} \\ \alpha_c &= 0.15 \end{aligned} \tag{12}$$

B. Phase Aware Dynamic Gait Correction

While the robot is actively ambulating, the correction pipeline switches to an adaptive PID controller specific to the gait cycle that incorporates three additional mechanisms:

Moving Average Baseline. IMU measurements are accumulated into a buffer spanning one full gait cycle. The running mean provides a smoothed estimate of the true tilt over the cycle, filtering out periodic oscillations inherent to legged locomotion.

Adaptive Proportional Gain. The proportional gain scales dynamically with the instantaneous tilt magnitude to provide stronger correction authority during large disturbances:

$$K_{P,eff} = K_P\left(1 + \min\left(1.0,\ \frac{\max(|\phi|, |\theta|)}{K_{prop}}\right)\right) \tag{13}$$

Zero Crossing Integral Reset. The integral accumulator is immediately reset to zero whenever the error signal crosses zero, preventing oscillatory overshoot during transient recovery.

Stance Phase Rotational Compensation. The corrective output is applied as a rotation matrix in the body frame $R = R_y(\theta)R_x(\phi)$:

$$R = \begin{bmatrix} c_\theta & s_\phi s_\theta & c_\phi s_\theta \\ 0 & c_\phi & -s_\phi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix} \tag{14}$$

This matrix is applied exclusively to limbs satisfying the stance condition $|Z_{foot} - Z_{stance}| \leq Z_{thresh}$, where $Z_{stance} = -170$ mm and $Z_{thresh} = 5$ mm. Only the vertical component of the transformed foot position is modified; horizontal coordinates are preserved to maintain stride geometry:

$$\mathbf{p}_{adj} = R \cdot \mathbf{p}_{raw} \tag{15}$$

$$x_{adj} = x_{raw}, \quad y_{adj} = y_{raw}$$

V. Results and Diagnostic Data

A comprehensive diagnostic pipeline continuously broadcasts internal controller states across a structured topic hierarchy to objectively quantify algorithmic performance during active testing. Data is presented separately for the two operating modes: static home posture and dynamic trotting gait.

A. Home Posture Stabilization

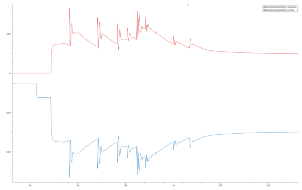
During the static home posture, the quadruped maintains all four limbs in ground contact while the PID controller actively rejects base orientation disturbances.

Fig. 4 presents the complete home mode diagnostic dataset. The pitch axis measurement and error signals (Fig. 4a) show the base tilting under an applied disturbance. The EMA filtered IMU measurement captures the orientation deviation, while the deadband filtered error signal demonstrates effective suppression of trivial oscillations below the ±0.02 rad threshold. The gradual convergence toward zero confirms the integral accumulator's correction capability at steady state.

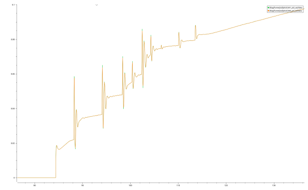
The corresponding PID correction output (Fig. 4b) reveals that the saturated PID output exhibits sharp transient spikes during rapid disturbance onset, while the lowpass filtered output tracks the saturated signal with significantly reduced high frequency content. The resulting joint torque commands (Fig. 4c) settle from initial transients into steady state values, confirming that the multistage pipeline produces stable actuator commands suitable for sustained static balancing.

B. Dynamic Trotting Gait Stabilization

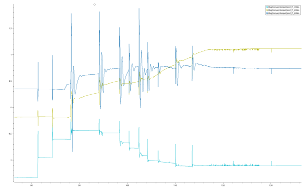
During active trotting locomotion, the adaptive PID controller specific to the gait cycle maintains postural stability while the quadruped executes cyclic quintic trajectories. Fig. 5 presents the complete trotting mode diagnostic dataset. The pitch axis measurement and error signals (Fig. 5a) exhibit characteristic periodic oscillations inherent to legged locomotion. The moving average error signal successfully extracts the sustained tilt trend from the cyclic gait artifacts, enabling the adaptive PID to distinguish genuine disturbances from normal locomotion



(a) *Pitch and error.*



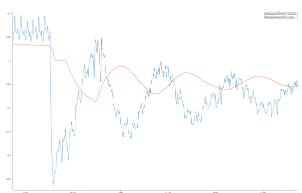
(b) *PID and LPF output.*



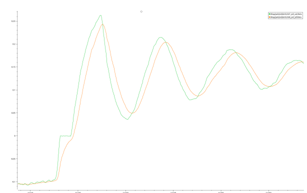
(c) *Left front joint torques.*

Fig. 4: *Home posture stabilization diagnostics. (a) Pitch measurement and deadband filtered error. (b) Saturated PID and lowpass filtered correction output. (c) Resulting joint torque commands for the left front limb.*

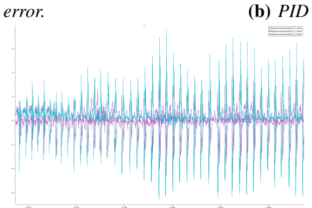
dynamics. The corresponding PID correction output (Fig. 5b) demonstrates the adaptive proportional gain responding to large tilt events with elevated correction authority, while the lowpass filtered output produces the smooth signal published to the gait generator for stance phase rotational compensation. The resulting joint torques (Fig. 5c) display cyclic profiles reflecting swing to stance phase transitions, with the stance phase rotational compensation visible as high frequency modulation in θ_3.



(a) *Pitch and error.*



(b) *PID and LPF output.*



(c) *Left front joint torques.*

Fig. 5: *Dynamic trotting gait stabilization diagnostics. (a) Pitch measurement and moving average error. (b) Adaptive PID and lowpass filtered correction output. (c) Resulting joint torque commands for the left front limb.*

VI. Conclusion and Future Work

This paper presented a decoupled dual pipeline control framework for quadruped locomotion that achieves robust postural stability through a novel multistage signal conditioning approach. The key contributions, namely quintic trajectory planning with enforced boundary conditions, gain scheduled PID with back calculation antiwindup, phase aware rotational stance compensation, and strict architectural isolation at the node level, collectively enable stable blind locomotion on embedded hardware without requiring computationally expensive MPC [2] or RL solvers.

The proposed multistage pipeline (EMA → deadband → gain scheduled PID → saturation → LPF) demonstrates that systematic cascading of conventional signal processing elements can produce corrective outputs of sufficient quality to stabilize complex mechanical systems with multiple articulated links, representing a practical alternative to monolithic optimal controllers.

Future work will pursue three principal directions:

1) Incorporation of Ground Reaction Force (GRF) estimation via motor current sensing, enabling terrain classification without dedicated sensors for force and torque measurement.
2) Extension of the gait library to incorporate adaptive stride modulation, dynamically adjusting L_{stride} and h_{lift} based on estimated terrain roughness.
3) Migration of the gain scheduled PID to an adaptive controller that is model free and automatically tunes K_P, K_I, K_D based on performance metrics from the closed loop.

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